

# Mohammad Huzaifa

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## Summary

B.Tech Computer Science student specializing in Agentic AI, Machine Learning, and Backend Engineering. Proven track record of designing autonomous AI agents, fine-tuning LLMs, and building robust data curation pipelines. Experienced in designing scalable backend systems, database schemas, and custom workflow automations to improve operational efficiency.

## Education

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|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>United Institute of Technology, Prayagraj</b><br><i>B.Tech in Computer Science and Engineering (7.47 CGPA up to 5th sem)</i> | 2023 – 2027 |
| <b>Allahabad Public School, Prayagraj</b><br><i>Intermediate (85.8%) &amp; High School (88.3%)</i>                              | 2020 – 2022 |

## Experience

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| <b>Sanfy Consultancy Services Pvt. Ltd. (Orvanto AI)</b> <u>Certificate</u><br><i>Backend Developer Intern</i>                                                                                                                                                                                                                                                                                                               | Apr 2026 – Jul 2026 |
| <ul style="list-style-type: none"><li>Designed and optimized high-throughput REST APIs and backend infrastructure using Python, FastAPI, and SQL.</li><li>Engineered custom n8n automation workflows and Docker environments to streamline data ingestion and system processes.</li><li>Collaborated with the product team to debug complex systems, integrate third-party APIs, and resolve database bottlenecks.</li></ul> |                     |

## Technical Projects

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| <b>AnveshakSutra — Zero-Knowledge Breach Intelligence</b> <u>GitHub</u> <u>Preview</u>                                                                                                                                                                                                                                                 | Aug 2026            |
| <ul style="list-style-type: none"><li>Architected a Zero-Knowledge K-Anonymity protocol (5-char SHA-256 prefix bucketing) querying breach pools with zero server identity leakage.</li><li>Built a 3D WebGL Graph ML blast radius engine (Betweenness Centrality) with Celery async sweepers and canary deception tripwires.</li></ul> |                     |
| <b>Concept3D — AI Concept-to-3D Visualization</b> <u>GitHub</u> <u>Preview</u>                                                                                                                                                                                                                                                         | Feb 2026 – Mar 2026 |
| <ul style="list-style-type: none"><li>Led backend and 3D data pipeline engineering to translate unstructured conceptual data into structured 3D generation inputs.</li><li>Streamlined generation workflows by designing an AI-driven semantic preprocessing pipeline (Python, PyTorch, Gen-AI).</li></ul>                             |                     |
| <b>VaakKavach — Autonomous Edge AI Acoustic Defense</b> <u>GitHub</u> <u>Preview</u>                                                                                                                                                                                                                                                   | Jan 2026 – Mar 2026 |
| <ul style="list-style-type: none"><li>Architected sub-15ms on-device acoustic DSP engine with native Kotlin AudioRecord ring buffers, FFT, and phase jitter metrics.</li><li>Engineered 100% offline serverless architecture with autonomous self-healing watchdog and SHA-256 SQLite forensic vault.</li></ul>                        |                     |
| <b>SunMap — 3D Spatial Solar Energy &amp; Intelligence Engine</b> <u>GitHub</u> <u>Preview</u>                                                                                                                                                                                                                                         | Oct 2025            |
| <ul style="list-style-type: none"><li>Architected a 60 FPS WebGL/Three.js spatial engine parsing CityGML LOD2 building geometries, normal vectors, and shadow occlusions.</li><li>Implemented Perez clear-sky transposition physics (NREL PVLlib benchmarked) with diurnal tracking across 8,760 annual vectors.</li></ul>             |                     |
| <b>KavachG — AI-Based Safety KPI &amp; Monitoring</b> <u>GitHub</u> <u>Preview</u>                                                                                                                                                                                                                                                     | Apr 2025 – Jun 2025 |
| <ul style="list-style-type: none"><li>Developed machine learning pipelines and backend service integrations for a smart safety monitoring dashboard.</li><li>Deployed real-time computer vision models to automate safety compliance checks (OpenCV, FastAPI).</li></ul>                                                               |                     |

## Skills

**Generative AI & Agentic Systems:** Agentic AI, LLM Fine-Tuning, RAG, Data Curation  
**Machine Learning & CV:** Machine Learning, Computer Vision, PyTorch, TensorFlow, Scikit-Learn, EDA  
**Backend & Data Engineering:** FastAPI, SQL (DBMS), ETL, Data Modeling, WebSocket API  
**Programming Languages:** Python, C++, C  
**Tools & Automation:** Docker, Git/GitHub, n8n (Workflow Automation), Figma (UI/UX)

## Publications & Certifications

- Physiochemical Pattern Fingerprinting (PPF): Protein Homology Detection (IJDDT, Vol. 16, Issue 4)**  
Co-authored a memory-efficient approach to structurally-sensitive homology detection.
- CS50's Introduction to Artificial Intelligence with Python** — Covered search, ML, NLP, and AI in Python.

## Achievements

- Prototype Sprint Qualifier, Sankalp 2026 (MNNIT Prayagraj)** — Built Concept3D for generating interactive 3D visualizations.
- Second Runner-Up, UHACK 4.0 Hackathon (United Group of Institutions)** — Built A.E.G.I.S audio scam detection pipeline.
- Second Runner-Up, CodeStorm'25 Hackathon (Shambhunath Group of Institutions)** — Built SunMap 3D CityGML spatial engine.